

MARWAN EL SAABI GONZÁLEZ

DATA SCIENCE & MACHINE LEARNING | MSC BIOINFORMATICS CANDIDATE

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PROFILE

Biologist turned data scientist, building ML systems end to end — from raw data to deployed, production-ready tools: LLM-powered applications, supervised ML, time-series forecasting and relational data modelling. Seeking a data science internship to keep shipping tools like these.

TECHNICAL PROJECTS & SOFTWARE DEVELOPMENT

AI-Driven Generative Application (De Novo Drug Designer) | Python, Streamlit, Google Gemini API, RDKit

- Architected and deployed an interactive open-source web application for *de novo* molecular generation and 3D rendering.
- Integrated the Gemini API with an RDKit validation layer to filter chemically invalid molecular structures.

Retail Demand Forecasting | Python, LightGBM, Pandas, Streamlit

- Forecast weekly SKU demand 4 weeks ahead from 1M+ transactions using LightGBM with lag, price and calendar features.
- Beat naive, seasonal-naive and moving-average baselines by 4.8% wMAPE under rolling-origin backtesting.

Alzheimer's Drug Classifier (ML Virtual Screening) | Python, Scikit-learn, XGBoost, RDKit

- Built a virtual screening pipeline on ChEMBL IC50 data, encoding molecules as 1024-bit Morgan fingerprints.
- Benchmarked RF, SVM, XGBoost and KNN under stratified 5-fold CV, validating the winner with leakage control.

RNA-seq Analysis Dashboard | Python, Streamlit, Plotly, Scikit-learn

- Built an interactive Streamlit dashboard for RNA-seq analysis: PCA, volcano/MA plots and heatmaps.
- Implemented differential expression analysis (Welch's t-test, BH FDR correction) on a public dataset (Himes et al., 2014).

Melanoma Detection (Classical Computer Vision) | Python, OpenCV, Scikit-learn, Scikit-image

- Automatic segmentation of dermoscopic lesions via Otsu, K-means and watershed, selected by IoU against ground truth.
- Extracted 40 shape, color and texture (LBP/GLCM) features; SVM reached 90.2% accuracy and 84% melanoma sensitivity.

PROFESSIONAL EXPERIENCE

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|---|------------------------|
| • Delivery Driver (Part-time) / Teigal Logística (Amazon Delivery Partner) | June 2026 to Present |
| • Production Operator / Stellantis España S.L, Vigo | Oct 2024 to Oct 2025 |
| • Seasonal Farmworker (Grape Harvest) / Iman Temporing ETT,S.L | Sept 2024 to Sept 2024 |

EDUCATION

University of A Coruña (Universidade da Coruña), A Coruña	Sept 2025 to Feb 2027
MSc in Bioinformatics Applied to Health Sciences	
University of Vigo (Universidade de Vigo), Vigo	Sept 2019 to Jun 2023
BSc in Biology (Specialization: Biotechnology)	

AREAS OF EXPERTISE

- **Programming & Data:** Python (NumPy, Pandas, OpenCV, Matplotlib), R (Tidyverse, ggplot2), SQL, MATLAB, Bash.
- **Machine Learning:** Scikit-learn, XGBoost, LightGBM, Neural Networks, SVMs, Cross-Validation, Time-Series Forecasting.
- **Tools & Infrastructure:** Linux, Git/GitHub, Docker, Conda, High-Performance Computing (SLURM).

LANGUAGES

English, Spanish, Galician

PUBLICATIONS & CERTIFICATIONS

Alarcón-Ríos L., **Saabi González M.**, Velo-Antón G. (2026). Melanism in salamanders: effects of insularity, sex and size on dorsal colouration in *Salamandra salamandra*. *Journal of Zoology*. [doi:10.1111/jzo.70133](https://doi.org/10.1111/jzo.70133)

English B2 – CertAcles (Universidade de Vigo), May 2025